

Pontryagin's Maximum Principle

Consider the optimal control problem of the form

$$\arg \inf_{u: [t, T] \rightarrow U} \left\{ G(z_T) + \int_t^T L(s, z_s, u_s) ds \mid z_t = x; \quad dz_s = f(s, z_s, u_s) ds, \quad t < s \leq T \right\}$$

where the Hamiltonian, defined as

$$H(t, x, p) = \sup_{u \in U} \{-p \cdot f(t, x, u) - L(t, x, u)\}$$

is assumed to admit a unique continuous closed-form solution $u^*(t, x, p)$, **Pontryagin's maximum principle** states that for a solution (z_s^*, u_s^*) of the control problem, there exists $p_s^* : [0, T] \rightarrow \mathbb{R}^d$ such that, for $t \leq s \leq T$,

$$dz_s^* = -\nabla_p H(s, z_s^*, p_s^*),$$

$$z_t^* = x,$$

$$dp_s^* = \nabla_x H(s, z_s^*, p_s^*),$$

$$p_T^* = \nabla G(z_T^*),$$

$$p_s^* \cdot f(s, z_s^*, u_s^*) + L(s, z_s^*, u_s^*) = -H(s, z_s^*, p_s^*), \quad \text{for all } s \in [t, T].$$

- ✓ grid-free (suitable for high-dimensional problems)
- × resulting control corresponds to fixed initial state (pathwise control)
- × forward-backward structure, non-convexity, hard to solve numerically

Dynamic Programming

Recall the classical optimal control problem

$$\arg \inf_{u: [t, T] \rightarrow U} \left\{ G(z_T) + \int_t^T L(s, z_s, u_s) ds \mid z_t = x; \quad dz_s = f(s, z_s, u_s) ds, \quad t < s \leq T \right\}$$

where the value function, defined as

$$\Phi(t, x) = \inf_u J_{x,t}[u] = \inf_u G(z_T) + \int_t^T L(s, z_s, u_s) ds$$

is the unique viscosity solution of the **Hamilton-Jacobi-Bellman equation**

$$-\partial_t \Phi(t, x) + H(t, x, \nabla_x \Phi(t, x)) = 0, \quad (t, x) \in [t, T] \times \mathbb{R}^d,$$

$$\Phi(T, x) = G(x).$$

Consequently, optimal control for initial data (t, x) and $s \in [t, T]$ reads

$$u_s^* \in \arg \min_{v \in U} \{ \nabla_x \Phi(s, z_s^*) \cdot f(s, z_s^*, v) + L(s, z_s^*, v) \}.$$

- ✓ can recover the optimal control at any state (**robust control**)
- ✓ necessary and sufficient conditions for optimality
- × curse of dimensionality, nonlinearity, **challenging to solve numerically**